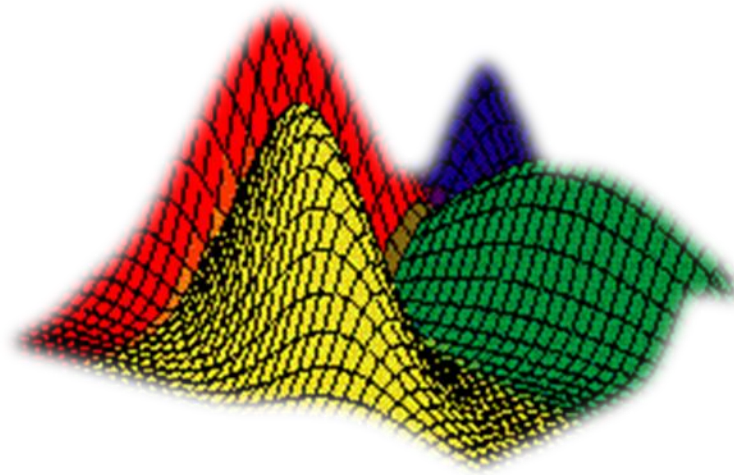
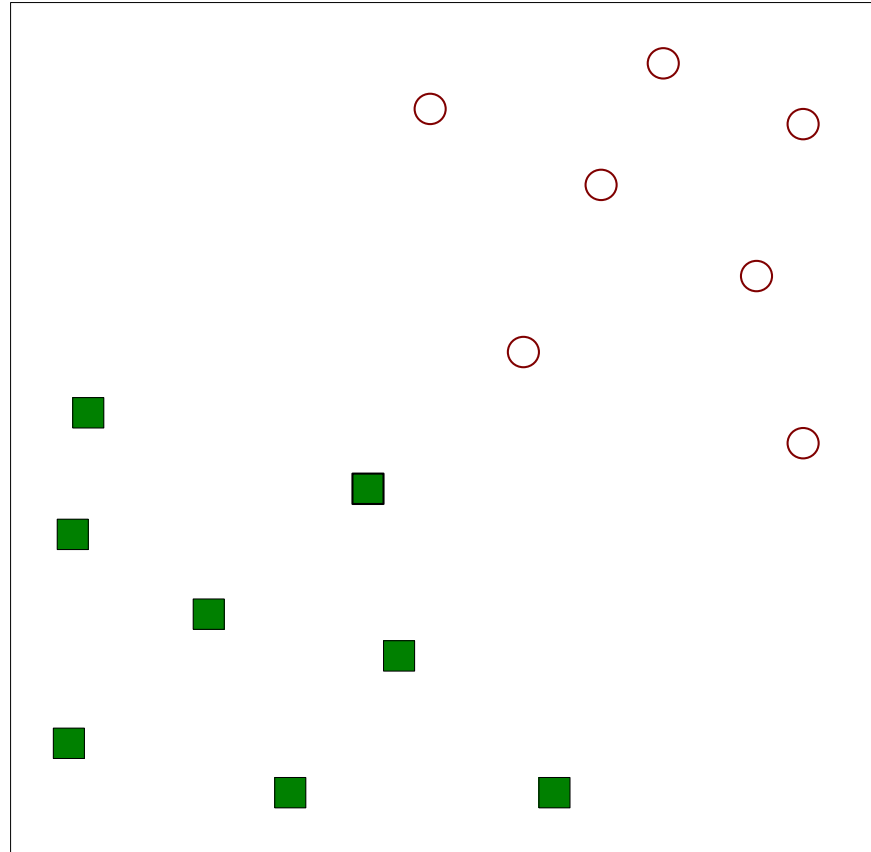




Pattern Recognition

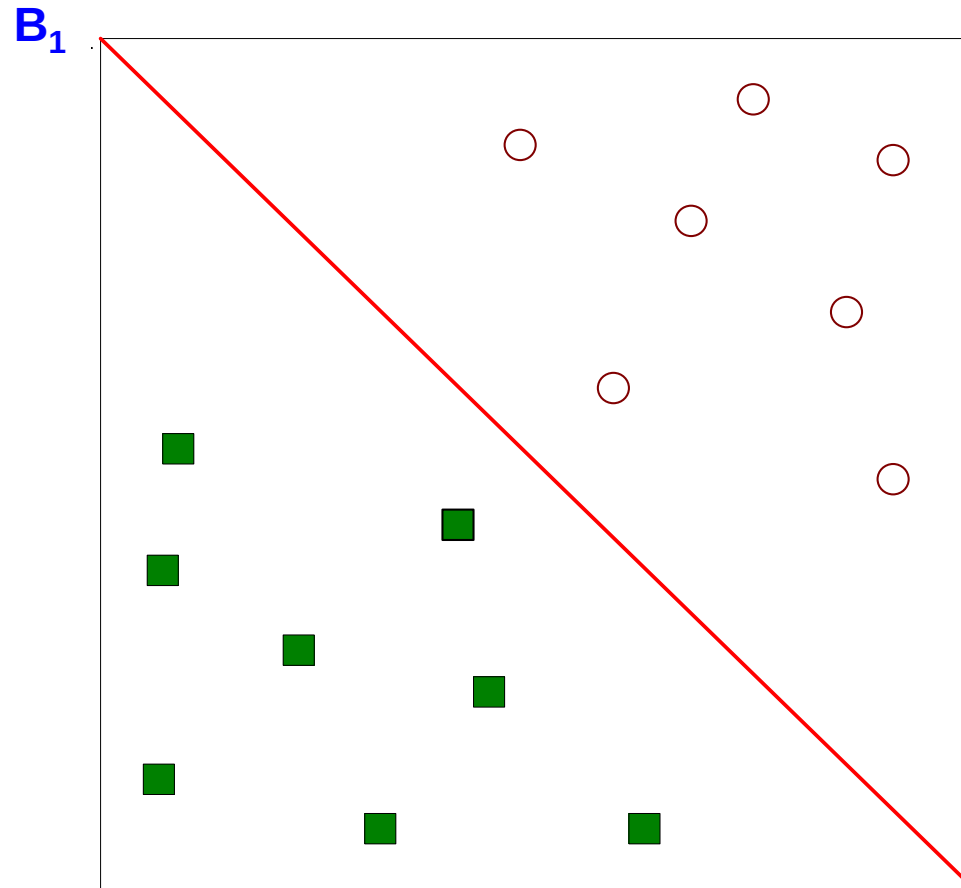
Linear Classifier

Two-Class case again



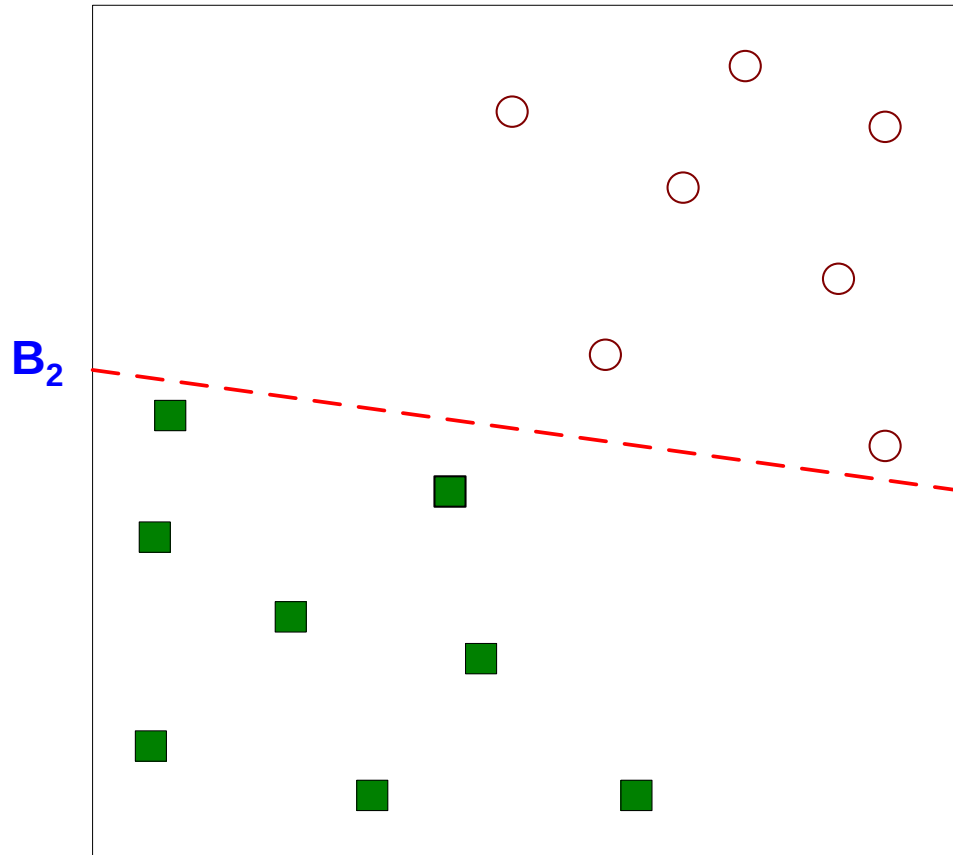
- Find a linear hyperplane (decision boundary) that will separate the data

Two-Class case again



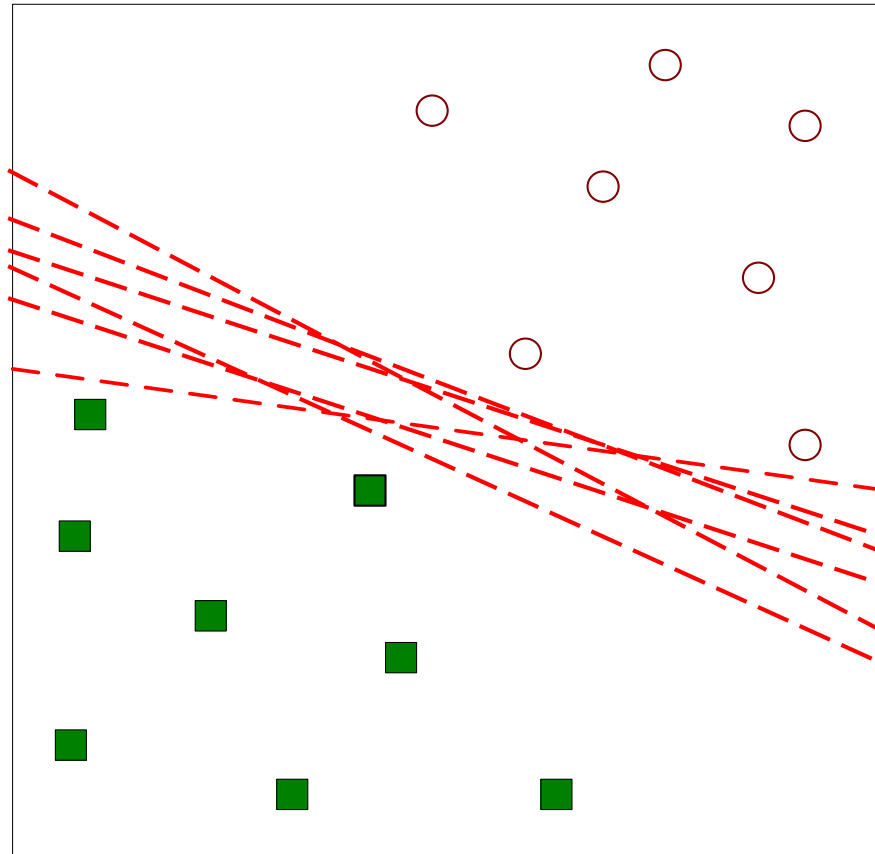
- One Possible Solution

Two-Class case again



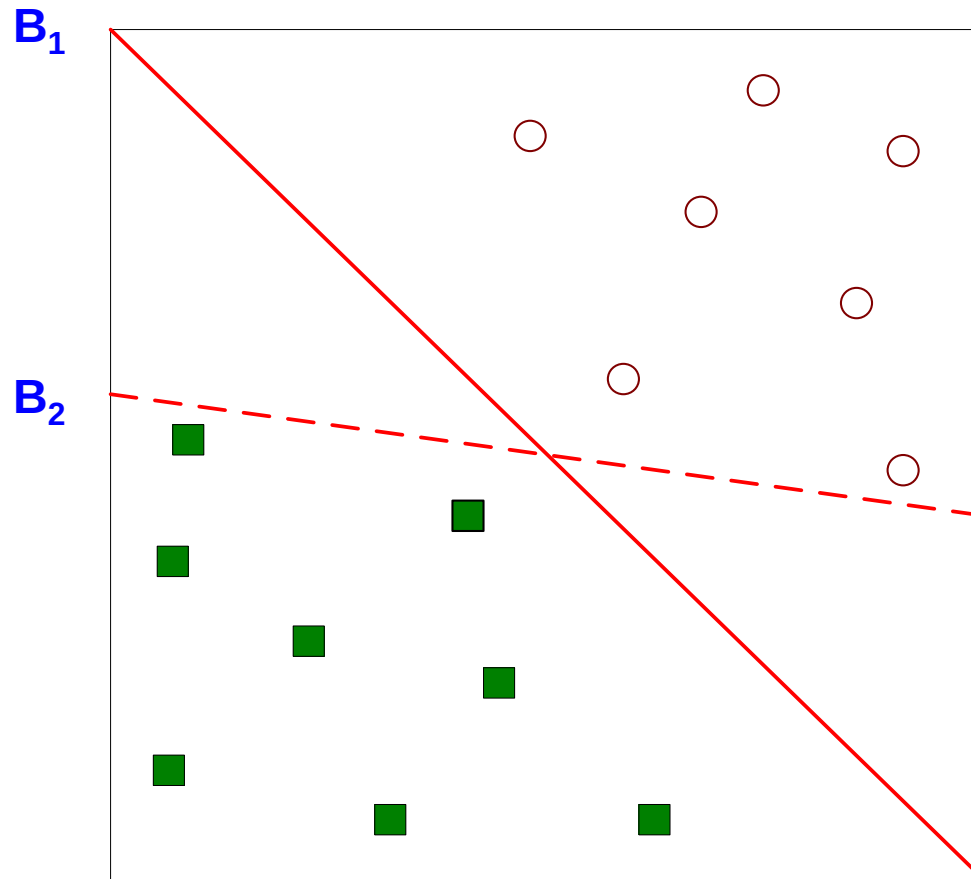
- Another possible solution

Two-Class case again



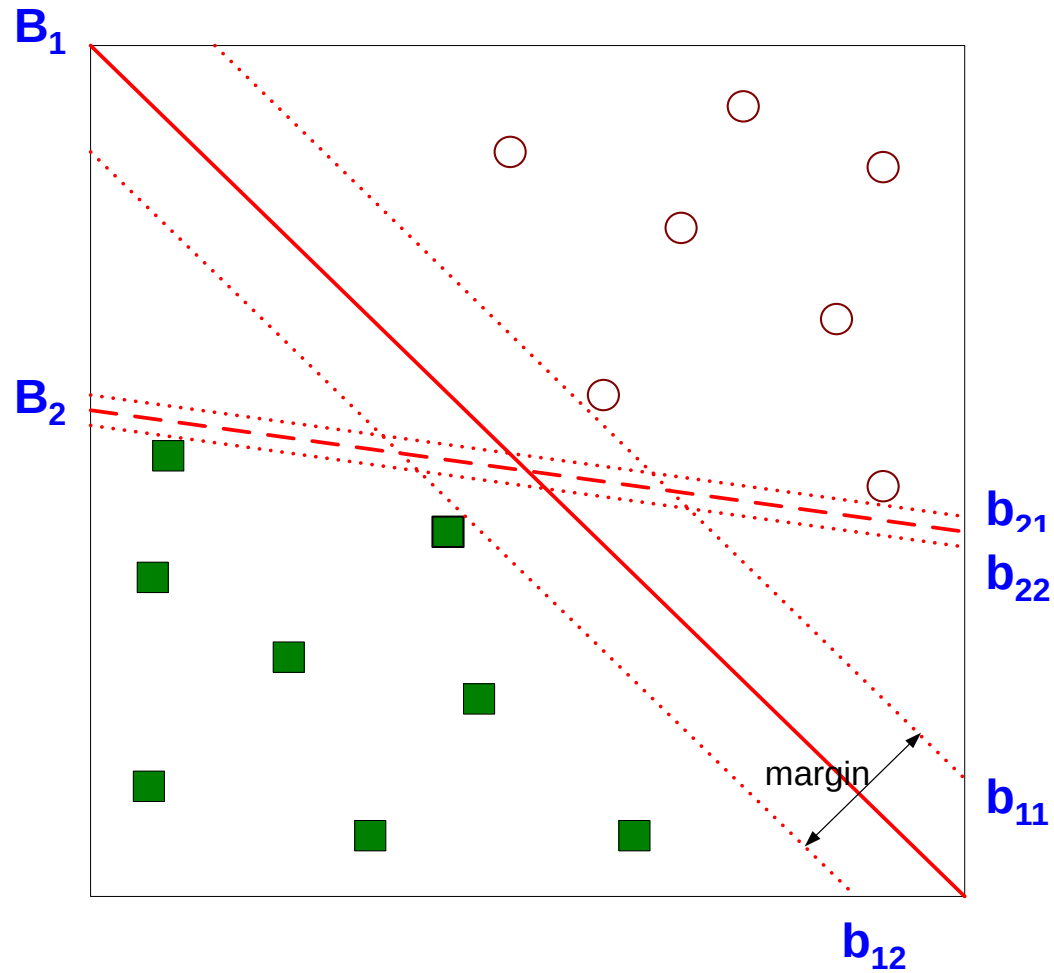
- Other possible solutions

Two-Class case again



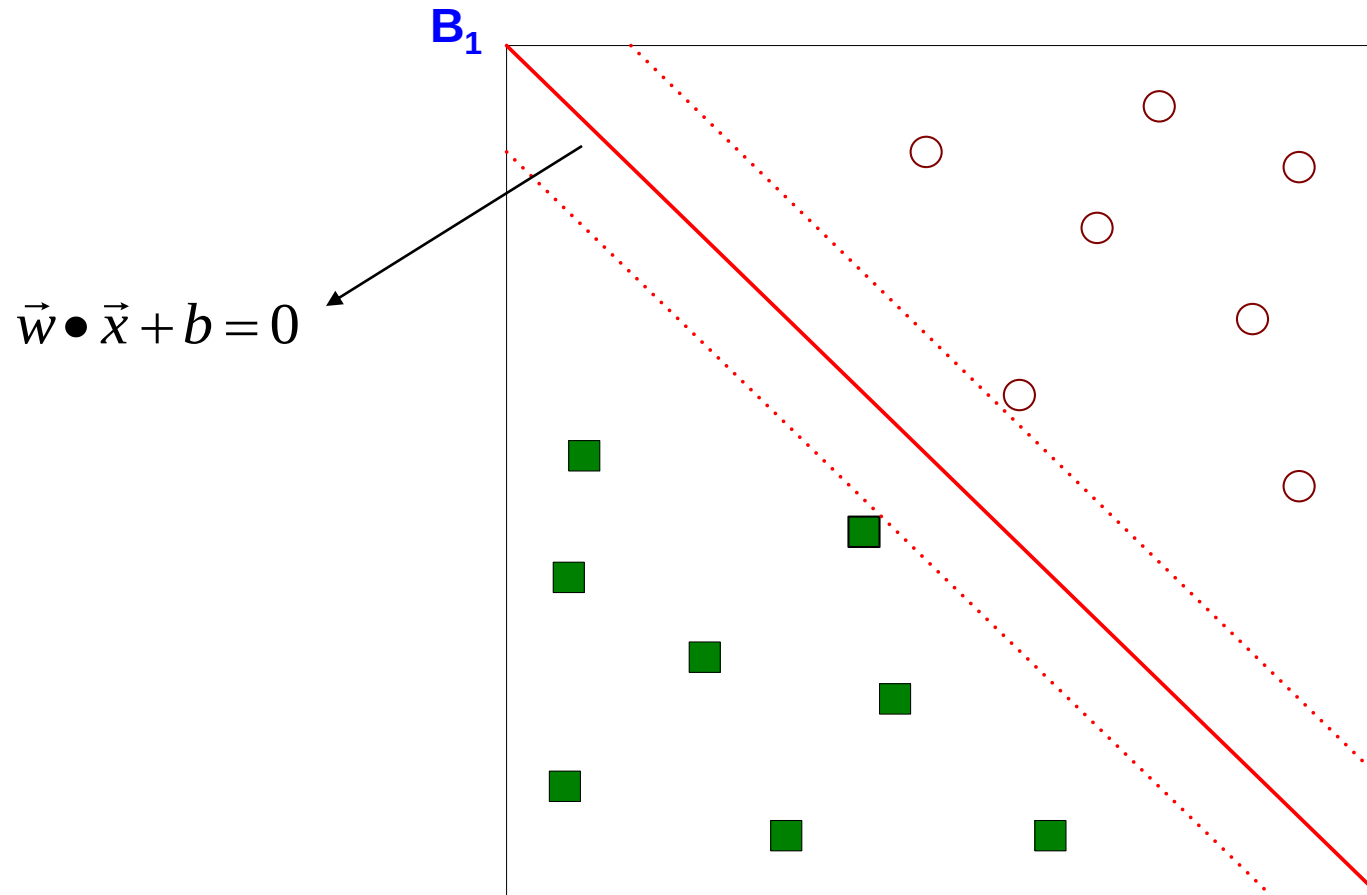
- Which one is better? B_1 or B_2 ?
- How do you define better?

Two-Class case again

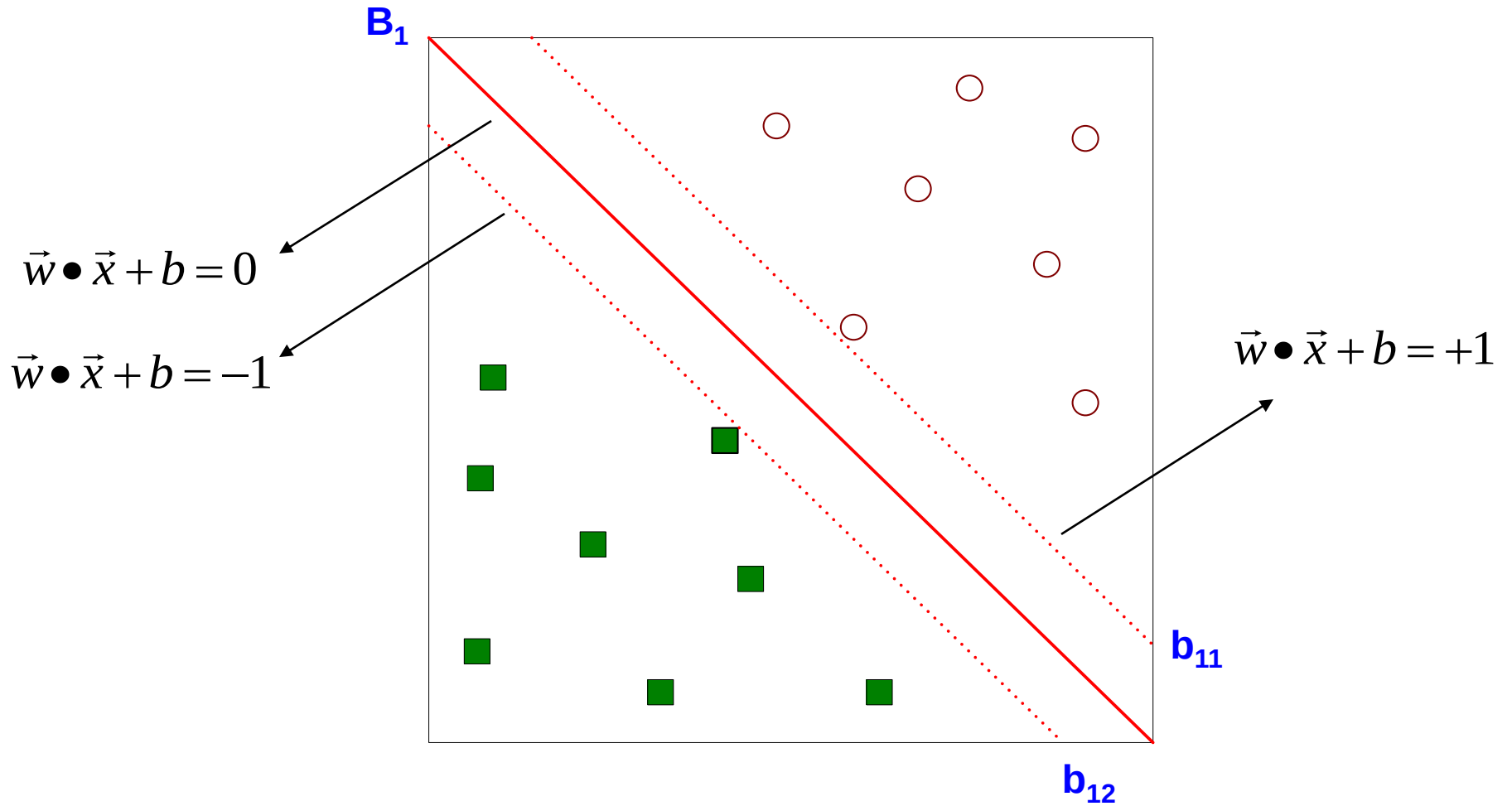


- Find hyperplane maximizes the margin $\Rightarrow B_1$ is better than B_2

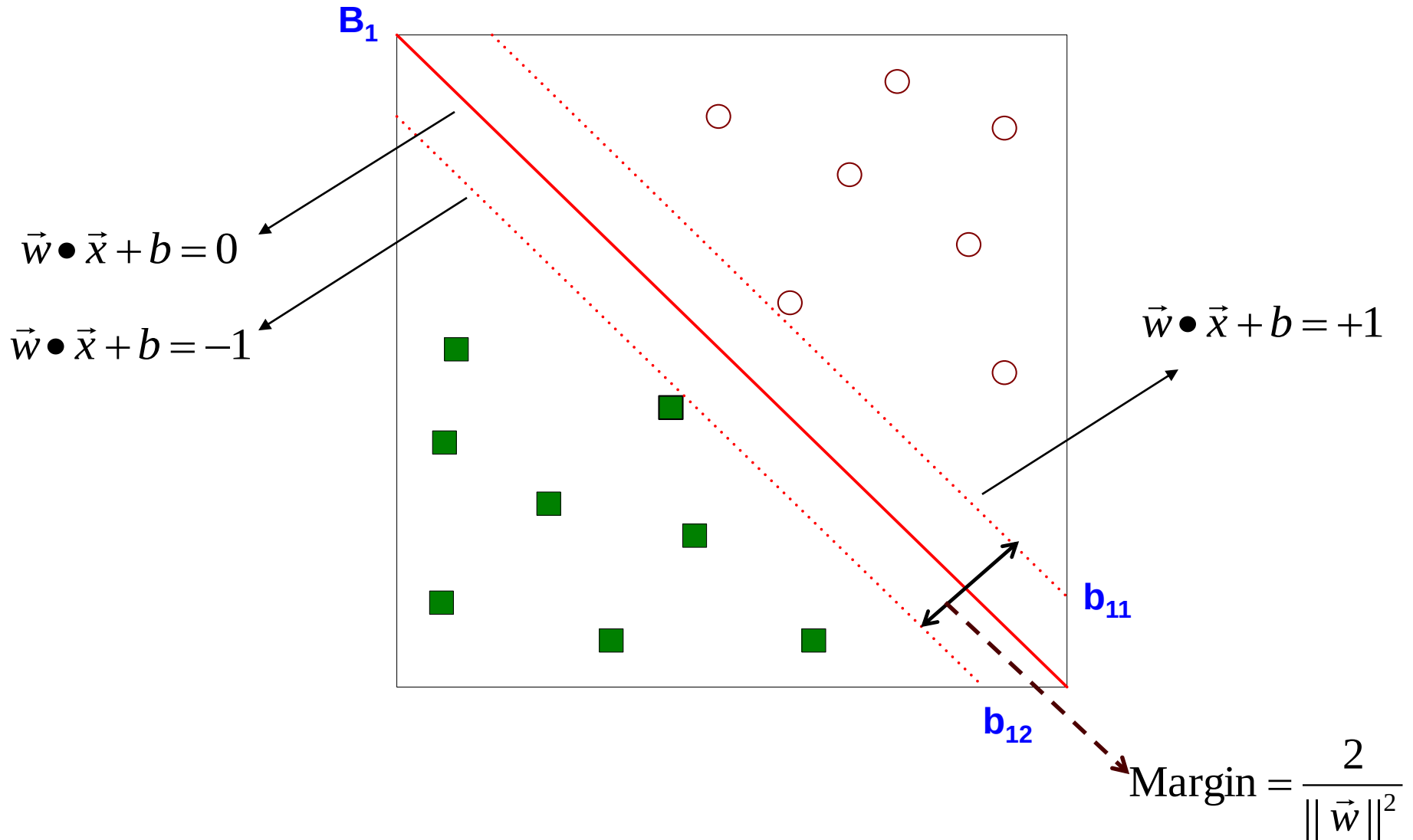
Two-Class case again



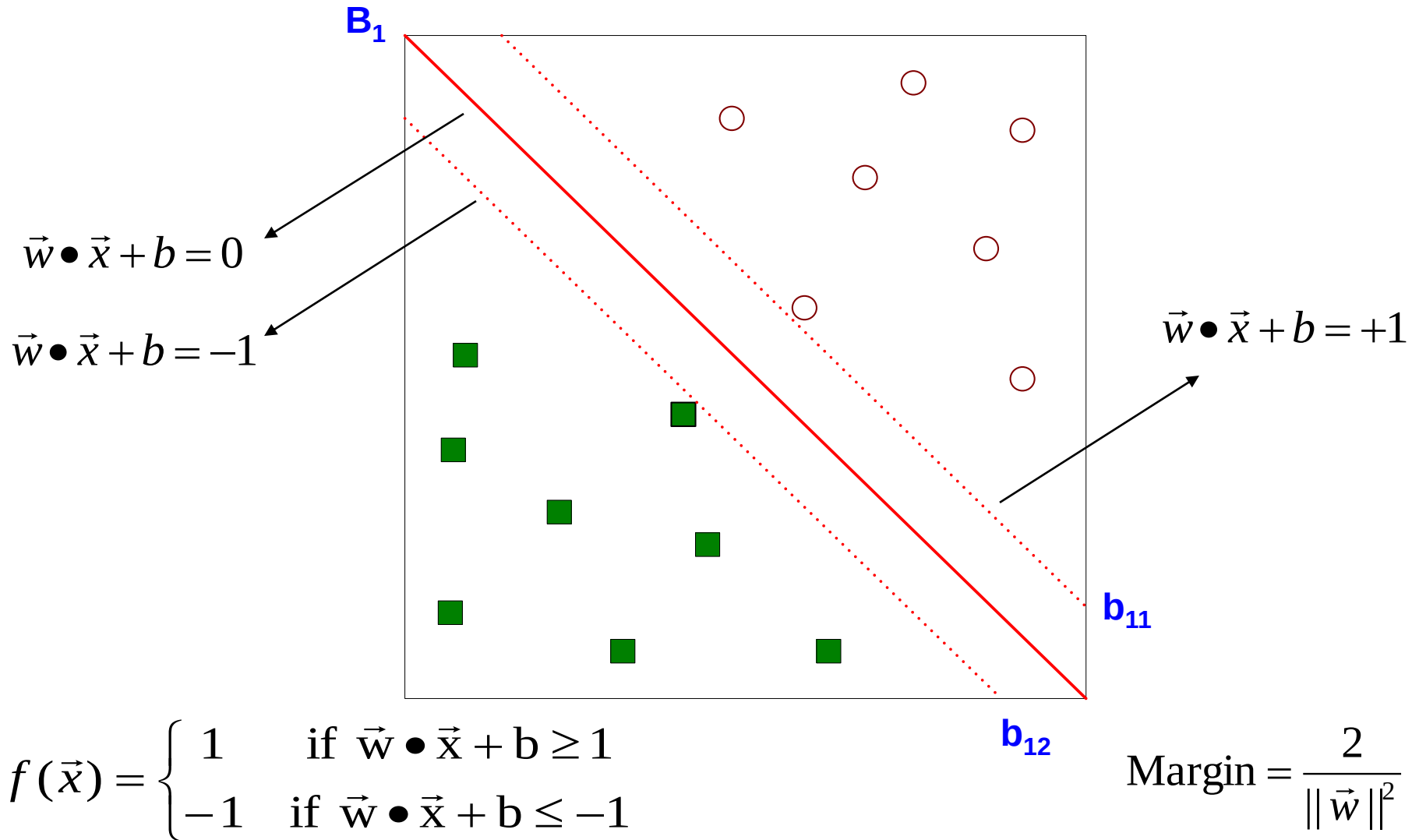
Two-Class case again



Two-Class case again



Two-Class case again



Support Vector Machines

- We want to maximize: $\text{Margin} = \frac{2}{\|\vec{w}\|^2}$

– But subjected to the following constraints:

$$y_i = \begin{cases} 1 & \text{if } \vec{w} \bullet \vec{x}_i + b \geq 1 \\ -1 & \text{if } \vec{w} \bullet \vec{x}_i + b \leq -1 \end{cases}$$

Support Vector Machines

- We want to maximize: $\text{Margin} = \frac{2}{\|\vec{w}\|^2}$

– Which is equivalent to minimizing:

$$L(w) = \frac{\|\vec{w}\|^2}{2}$$

– But subjected to the following constraints:

- $y_i = \begin{cases} 1 & \text{if } \vec{w} \bullet \vec{x}_i + b \geq 1 \\ -1 & \text{if } \vec{w} \bullet \vec{x}_i + b \leq -1 \end{cases}$

Support Vector Machines

The Expression

$$y_i = \begin{cases} 1 & \text{if } \vec{w} \bullet \vec{x}_i + b \geq 1 \\ -1 & \text{if } \vec{w} \bullet \vec{x}_i + b \leq -1 \end{cases}$$

can be written as

$$y_i (\vec{w} \bullet \vec{x}_i + b) \geq 1$$

Support Vector Machines

The Expression

$$y_i = \begin{cases} 1 & \text{if } \vec{w} \bullet \vec{x}_i + b \geq 1 \\ -1 & \text{if } \vec{w} \bullet \vec{x}_i + b \leq -1 \end{cases}$$

can be written as

$$y_i (\vec{w} \bullet \vec{x}_i + b) \geq 1$$

- We can say :

- minimize:

$$L(w) = \frac{\|\vec{w}\|^2}{2}$$

- Subject to:

$$y_i (\vec{w} \bullet \vec{x}_i + b) \geq 1$$

Support Vector Machines

- $L(\mathbf{w}) = \frac{\|\vec{\mathbf{w}}\|^2}{2}$; a quadratic equation
- Solving for $\mathbf{w}$ and $\mathbf{b}$ is not easy
- What happens if $\mathbf{w}$ = 0?

Support Vector Machines

- minimize: $L(w) = \frac{\|\vec{w}\|^2}{2}$
- Subject to: $y_i(\vec{w} \bullet \vec{x}_i + b) \geq 1$

- Use Lagrange function:

$$L_p = \frac{\|\vec{w}\|^2}{2} + \sum_{i=1}^N \lambda_i (y_i(w \cdot x_i + b) - 1)$$

Support Vector Machines

- Lagrange function:

$$L_p = \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1)$$

- New constraints are:

$$\frac{\partial L_p}{\partial \vec{w}} = 0$$

$$\frac{\partial L_p}{\partial b} = 0$$

Support Vector Machines

- Lagrange function:

$$L_p = \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1)$$

- New constraints are:

$$\frac{\partial L_p}{\partial \vec{w}} = 0 \quad \Rightarrow \quad \vec{w} = \sum_{i=1}^N \lambda_i y_i \vec{x}_i$$

$$\frac{\partial L_p}{\partial b} = 0 \quad \Rightarrow \quad \sum_{i=1}^N \lambda_i y_i = 0$$

Support Vector Machines

- Lagrange function:

$$L_p = \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1)$$

- constraints are:

$$\vec{w} = \sum_{i=1}^N \lambda_i y_i \vec{x}_i$$

Still not solvable, many variables

$$\sum_{i=1}^N \lambda_i y_i = 0$$

Support Vector Machines

- Use Lagrange function:

$$L_p = \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1)$$

- constraints are:

$$\vec{w} = \sum_{i=1}^N \lambda_i y_i \vec{x}_i$$

$$\sum_{i=1}^N \lambda_i y_i = 0$$

From Karush-Kuhn_Tucker Transform,

$$\lambda_i \geq 0$$

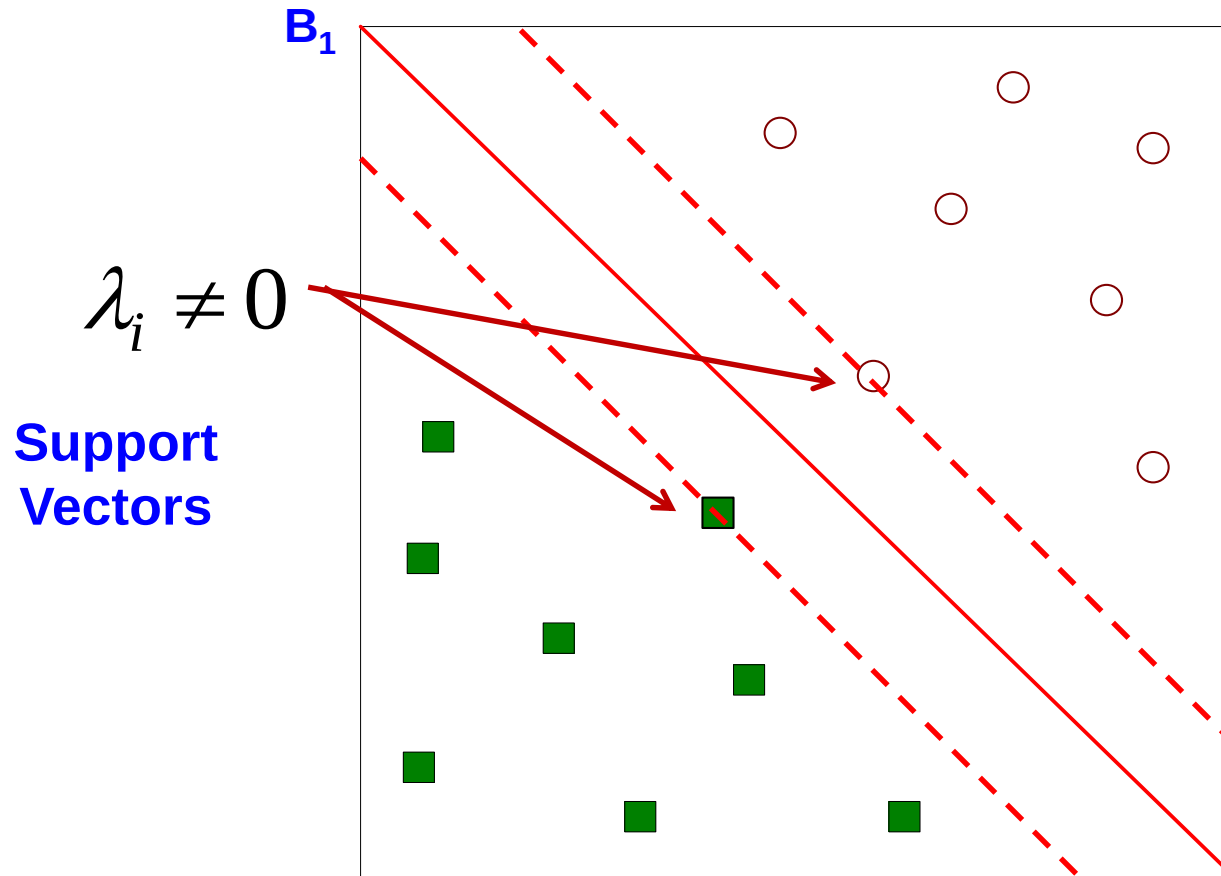
$$\lambda_i [y_i (\vec{w} \cdot \vec{x}_i + b) - 1] = 0$$

Support Vector Machines

$\lambda_i \geq 0$: non - negative

$$\lambda_i [y_i (\vec{w} \cdot \vec{x}_i + b) - 1] = 0$$

Support Vector Machines



$$\lambda_i \geq 0$$

$$\lambda_i [y_i (\vec{w} \cdot \vec{x}_i + b) - 1] = 0$$

Support Vector Machines

- Replace w with λ 's in L_p :

put $\vec{w} = \sum_{i=1}^N \lambda_i y_i \vec{x}_i$ and $\sum_{i=1}^N \lambda_i y_i = 0$

in $L_p = \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1)$

$$L_p = \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1)$$

$$\begin{aligned}
 L_p &= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1) \\
 &= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i y_i \vec{w} \cdot \vec{x}_i - \sum_{i=1}^N \lambda_i y_i b + \sum_{i=1}^N \lambda_i
 \end{aligned}$$

$$\begin{aligned}
L_p &= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1) \\
&= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i y_i \vec{w} \cdot \vec{x}_i - \sum_{i=1}^N \lambda_i y_i b + \sum_{i=1}^N \lambda_i \\
&= \frac{\|\vec{w}\|^2}{2} - \vec{w} \cdot \sum_{i=1}^N \lambda_i y_i \vec{x}_i - b \sum_{i=1}^N \lambda_i y_i + \sum_{i=1}^N \lambda_i
\end{aligned}$$

$$\begin{aligned}
L_p &= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1) \\
&= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i y_i \vec{w} \cdot \vec{x}_i - \sum_{i=1}^N \lambda_i y_i b + \sum_{i=1}^N \lambda_i \\
&= \frac{\|\vec{w}\|^2}{2} - \vec{w} \cdot \sum_{i=1}^N \lambda_i y_i \vec{x}_i - b \sum_{i=1}^N \lambda_i y_i + \sum_{i=1}^N \lambda_i \\
&= \frac{\|\vec{w}\|^2}{2} - \vec{w} \cdot \vec{w} - b \times 0 + \sum_{i=1}^N \lambda_i
\end{aligned}$$

$$\vec{w} = \sum_{i=1}^N \lambda_i y_i \vec{x}_i$$

$$\sum_{i=1}^N \lambda_i y_i = 0$$

$$\begin{aligned}
L_p &= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1) \\
&= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i y_i \vec{w} \cdot \vec{x}_i - \sum_{i=1}^N \lambda_i y_i b + \sum_{i=1}^N \lambda_i \\
&= \frac{\|\vec{w}\|^2}{2} - \vec{w} \cdot \sum_{i=1}^N \lambda_i y_i \vec{x}_i - b \sum_{i=1}^N \lambda_i y_i + \sum_{i=1}^N \lambda_i \\
&= \frac{\|\vec{w}\|^2}{2} - \vec{w} \cdot \vec{w} - b \times 0 + \sum_{i=1}^N \lambda_i \\
&= \sum_{i=1}^N \lambda_i + \frac{\vec{w} \cdot \vec{w}}{2} - \vec{w} \cdot \vec{w}
\end{aligned}$$

$$\begin{aligned}
L_p &= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1) \\
&= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i y_i \vec{w} \cdot \vec{x}_i - \sum_{i=1}^N \lambda_i y_i b + \sum_{i=1}^N \lambda_i \\
&= \frac{\|\vec{w}\|^2}{2} - \vec{w} \cdot \sum_{i=1}^N \lambda_i y_i \vec{x}_i - b \sum_{i=1}^N \lambda_i y_i + \sum_{i=1}^N \lambda_i \\
&= \frac{\|\vec{w}\|^2}{2} - \vec{w} \cdot \vec{w} - b \times 0 + \sum_{i=1}^N \lambda_i \\
&= \sum_{i=1}^N \lambda_i + \frac{\vec{w} \cdot \vec{w}}{2} - \vec{w} \cdot \vec{w} \\
&= \sum_{i=1}^N \lambda_i - \frac{\vec{w} \cdot \vec{w}}{2}
\end{aligned}$$

$$\begin{aligned}
L_p &= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1) \\
&= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i y_i \vec{w} \cdot \vec{x}_i - \sum_{i=1}^N \lambda_i y_i b + \sum_{i=1}^N \lambda_i \\
&= \frac{\|\vec{w}\|^2}{2} - \vec{w} \cdot \sum_{i=1}^N \lambda_i y_i \vec{x}_i - b \sum_{i=1}^N \lambda_i y_i + \sum_{i=1}^N \lambda_i
\end{aligned}$$

.

.

$$\begin{aligned}
&= \sum_{i=1}^N \lambda_i - \frac{\vec{w} \cdot \vec{w}}{2} \\
&= \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i=1}^N \lambda_i y_i \vec{x}_i \cdot \sum_{j=1}^N \lambda_j y_j \vec{x}_j
\end{aligned}$$

$$\begin{aligned}
L_p &= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1) \\
&= \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i y_i \vec{w} \cdot \vec{x}_i - \sum_{i=1}^N \lambda_i y_i b + \sum_{i=1}^N \lambda_i \\
&= \frac{\|\vec{w}\|^2}{2} - \vec{w} \cdot \sum_{i=1}^N \lambda_i y_i \vec{x}_i - b \sum_{i=1}^N \lambda_i y_i + \sum_{i=1}^N \lambda_i
\end{aligned}$$

.

.

$$\begin{aligned}
&= \sum_{i=1}^N \lambda_i - \frac{\vec{w} \cdot \vec{w}}{2} \\
&= \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i=1}^N \lambda_i y_i \vec{x}_i \cdot \sum_{j=1}^N \lambda_j y_j \vec{x}_j \\
&= \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j y_i y_j \vec{x}_i \cdot \vec{x}_j
\end{aligned}$$

Support Vector Machines

- Replace w with λ 's in L_p :

$$L_p = \frac{\|\vec{w}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{w} \cdot \vec{x}_i + b) - 1)$$

- The dual to be maximized:

$$L_D = \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j y_i y_j \mathbf{x}_i \cdot \mathbf{x}_j$$

Support Vector Machines

- After solving λ 's :
 - Find $\mathbf{w}$ and b :

$$\frac{\partial L_p}{\partial \mathbf{w}} = 0 \quad \Rightarrow \quad \mathbf{w} = \sum_{i=1}^N \lambda_i y_i \mathbf{x}_i$$

$$\vec{\mathbf{w}} \bullet \vec{\mathbf{x}}_i + b = 1$$

Support Vector Machines

- Classify an unknown example $\underline{\mathbf{z}}$:

$$f(\mathbf{z}) = \text{sign}(\mathbf{w} \cdot \mathbf{z} + b)$$

Table 1: Training Data

A_1	A_2	y	λ_i
0.38	0.47	+	65.52
0.49	0.61	-	65.52
0.92	0.41	-	0
0.74	0.89	-	0
0.18	0.58	+	0
0.41	0.35	+	0
0.93	0.81	-	0
0.21	0.10	+	0

Illustration : Linear SVM

- Consider the case of a binary classification starting with a training data of 8 tuples as shown in Table 1.
- Using quadratic programming, we can solve the KKT constraints to obtain the Lagrange multipliers λ_i for each training tuple, which is shown in Table 1.
- Note that only the first two tuples are support vectors in this case.
- Let $W = (w_1, w_2)$ and b denote the parameter to be determined now. We can solve for w_1 and w_2 as follows:

$$w_1 = \sum_i \lambda_i \cdot y_i \cdot x_{i1} = 65.52 \times 1 \times 0.38 + 65.52 \times -1 \times 0.49 = -6.64 \quad (22)$$

$$w_2 = \sum_i \lambda_i \cdot y_i \cdot x_{i2} = 65.52 \times 1 \times 0.47 + 65.52 \times -1 \times 0.61 = -9.32$$

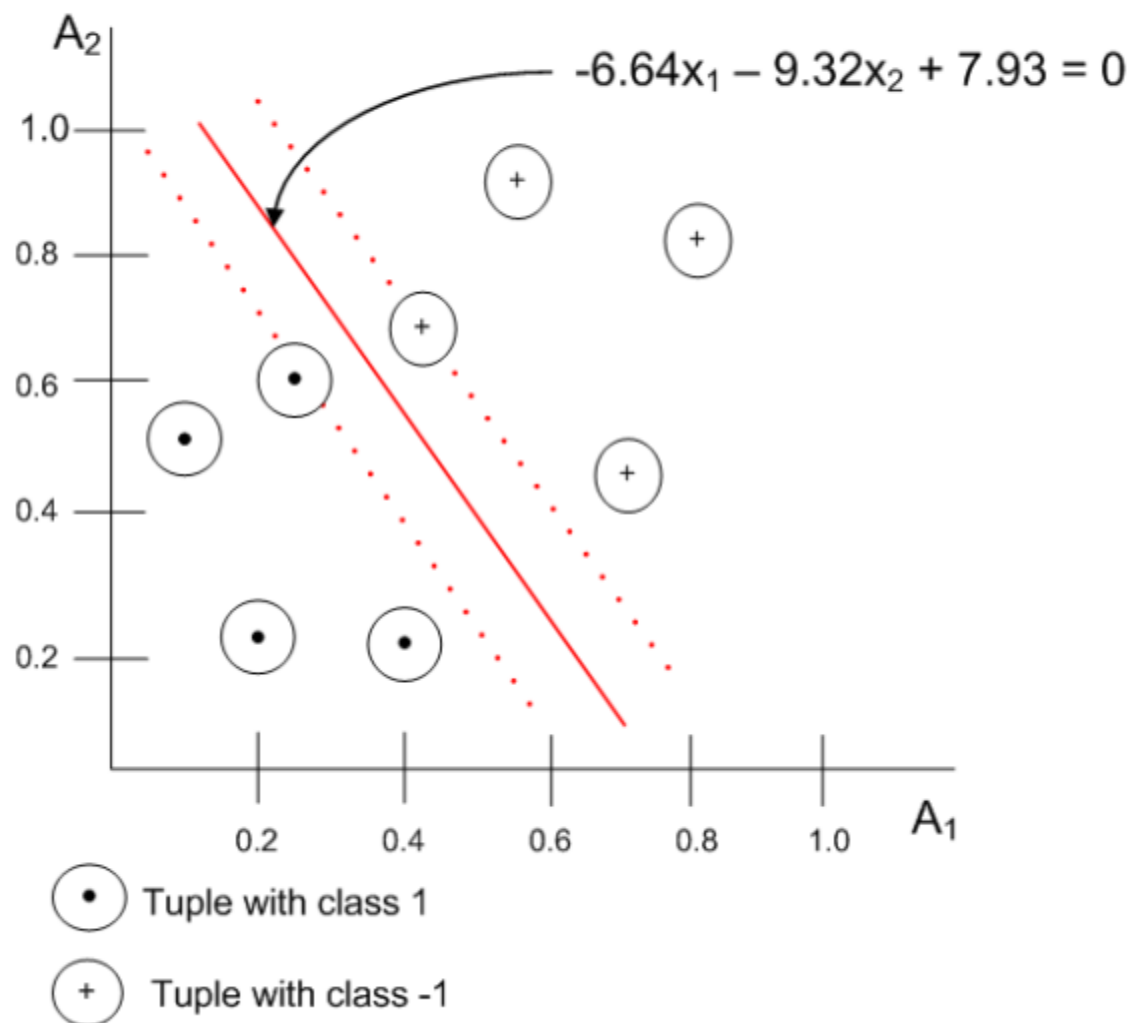
- The parameter b can be calculated for each support vector as follows

$$\begin{aligned} b_1 &= 1 - W \cdot x_1 \text{ // for support vector } x_1 \\ &= 1 - (-6.64) \times 0.38 - (-9.32) \times 0.47 \text{ //using dot product} \\ &= 7.93 \end{aligned}$$

$$\begin{aligned} b_2 &= 1 - W \cdot x_2 \text{ // for support vector } x_2 \\ &= 1 - (-6.64) \times 0.48 - (-9.32) \times 0.611 \text{ //using dot product} \\ &= 7.93 \end{aligned}$$

- Averaging these values of b_1 and b_2 , we get $b = 7.93$.

Figure 6: Linear SVM example.



- Thus, the MMH is $-6.64x_1 - 9.32x_2 + 7.93 = 0$ (also see Fig. 6).
- Suppose, test data is $X = (0.5, 0.5)$. Therefore,
$$\begin{aligned}\delta(X) &= W.X + b \\ &= -6.64 \times 0.5 - 9.32 \times 0.5 + 7.93 \\ &= -0.05 \\ &= -ve\end{aligned}$$
- This implies that the test data falls on or below the MMH and SVM classifies that X belongs to class label -.

Support Vector Machines

- What if the problem is not linearly separable?

